

Vigenère Cipher Practice Tasks

Task 1: Basic Encryption

Encrypt the message “SECURITYALERT” using the keyword “LOCK”. * Show your step-by-step work
* Include a table showing the calculation for each letter * Mark any calculations where you needed to subtract 26

Task 2: Mixed Case and Spaces

Encrypt the message “The Safe is Open” using the keyword “VAULT”. Requirements: * Maintain the original capitalization * Keep spaces in the encrypted text * Show how you handle spaces in the key repetition

Task 3: Decryption Challenge

Given the encrypted message “KPXVA QICKR ZTROI” and knowing it was encrypted using the keyword “GOLD”, decrypt the original message. * Show your step-by-step decryption process * Explain how you perform subtraction instead of addition for decryption * Present your work in a clear tabular format

Additional Requirements for All Tasks:

1. Use the standard alphabet where A=1, B=2, ..., Z=26
2. Show all mathematical calculations
3. Circle or highlight any cases where you need to subtract 26
4. Verify your final answer by encrypting it again to get the original ciphertext

Submission Format:

- Present all work neatly and organized
- Include tables for calculations
- Mark clearly where the key repeats
- Show final answers in both upper and lowercase formats